

1 Foundations for the convergence proof

We first establish two facts used throughout the proof. A policy that chooses actions at least as good as those of a reference policy improves its values. Also, sampled updates remain bounded and converge whenever their policy-evaluation target eventually stops changing. The second fact needs infinite accumulated learning at each pair, rather than a positive lower bound on its sampling probability.

1.1 Values and policy improvement

Let $\mathcal{S}$ be a finite set of nonterminal states, with finite nonempty action sets $\mathcal{A}(s)$, and write

$$\mathcal{I} = \{(s, a) : s \in \mathcal{S}, a \in \mathcal{A}(s)\}, \quad J = |\mathcal{I}|.$$

Terminal states have value zero and are omitted from $\mathcal{I}$. One-step rewards have finite expected absolute value. The set $\mathcal{P}$ consists of deterministic stationary policies. We assume either $0 \leq \gamma < 1$, or $\gamma = 1$ and every policy in $\mathcal{P}$ reaches a terminal state almost surely from every state. Such a policy is called proper.

The state value $v_\pi(s)$ is the expected discounted return starting at s and following π . The action value $q_\pi(s, a)$ prescribes the initial action a and follows π thereafter. Let $p(s' | s, a)$ be the transition probability and $r(s, a)$ the expected immediate reward. Then

$$q_\pi(s, a) = r(s, a) + \gamma \sum_{s' \in \mathcal{S}} p(s' | s, a) v_\pi(s'), \quad v_\pi(s) = q_\pi(s, \pi(s)). \quad (1)$$

All vector inequalities are coordinatewise and $\|\cdot\|_\infty$ denotes the maximum absolute coordinate. Policy order $\pi' \succeq \pi$ means $v_{\pi'} \geq v_\pi$, which implies $q_{\pi'} \geq q_\pi$ by (1). The converse need not hold: taking a prescribed initial action can bypass the only decision at which the policies differ.

We record why the episodic assumption suffices even when comparing a stationary policy with arbitrary control laws.

Lemma 1 (Uniform termination) *If every deterministic stationary policy is proper, there exist an integer $m \geq 1$ and $\rho < 1$ such that the termination time T satisfies $\mathbb{P}(T > jm) \leq \rho^j$ for every $j \geq 0$, uniformly over initial states, prescribed initial actions and history-dependent control laws. In particular, $\sup \mathbb{E}[T^2] < \infty$.*

Proof. Let $w_n(s)$ be the largest probability of remaining nonterminal after n transitions. Conditioning on the first transition gives

$$w_0(s) = 1, \quad w_{n+1}(s) = \max_{a \in \mathcal{A}(s)} \sum_{s'} p(s' | s, a) w_n(s').$$

These probabilities decrease to a vector w . Since the sets are finite, the limit satisfies the same equation. Choose a stationary maximizing policy π in this equation, and denote its nonterminal transition matrix by P_π . Then $w = P_\pi w = P_\pi^n w$. The row sum $\sum_{s'} P_\pi^n(s, s')$ is the probability of remaining nonterminal after n transitions starting at s . Properness makes each such probability tend to zero. Hence every entry of P_π^n tends to zero, and $w = 0$. Choose m with $\max_s w_m(s) \leq \rho < 1$. Conditioning after each block of m transitions proves

the bound. Prescribing the first action cannot increase the maximum. For an integer-valued $T \geq 0$, the identity $T^2 = \sum_{n=0}^{T-1} (2n+1)$ gives $\mathbb{E}[T^2] = \sum_{n \geq 0} (2n+1) \mathbb{P}(T > n)$. The geometric bound makes this sum finite uniformly over the control law. ■

Values are finite under either discounting or properness. Since $\mathcal{P}$ is finite, so is

$$B = \max_{\pi \in \mathcal{P}} \|q_\pi\|_\infty. \quad (2)$$

Lemma 2 (Policy improvement) *If $\pi, \pi' \in \mathcal{P}$ satisfy*

$$q_\pi(s, \pi'(s)) \geq q_\pi(s, \pi(s)) \quad (s \in \mathcal{S}), \quad (3)$$

then $v_{\pi'} \geq v_\pi$ and $q_{\pi'} \geq q_\pi$. If $q_{\pi'} \neq q_\pi$, both vector comparisons are strict in at least one coordinate. A policy greedy for its own action values is optimal, including among history-dependent control laws. An optimal deterministic stationary policy π_ exists.*

Proof. Write $r_{\pi'}(s) = r(s, \pi'(s))$ and $P_{\pi'}(s, s') = p(s' | s, \pi'(s))$. Set

$$h = r_{\pi'} + \gamma P_{\pi'} v_\pi - v_\pi \geq 0.$$

Subtracting the value equations gives $v_{\pi'} - v_\pi = h + \gamma P_{\pi'}(v_{\pi'} - v_\pi)$. Substituting this identity n times yields the first n terms below plus the remainder $(\gamma P_{\pi'})^n(v_{\pi'} - v_\pi)$. The remainder tends to zero: discounting bounds the row sums by γ^n , while properness makes the nonterminal transition probabilities tend to zero. Thus

$$v_{\pi'} - v_\pi = \sum_{j=0}^{\infty} (\gamma P_{\pi'})^j h \geq 0. \quad (4)$$

Equation (1) gives the action-value comparison. If the state values were equal, that equation would make the action values equal as well, proving the strictness assertion.

If π is greedy for q_π , then for every action a ,

$$v_\pi(s) \geq r(s, a) + \gamma \sum_{s'} p(s' | s, a) v_\pi(s').$$

Write s_t and r_t for its trajectory state and realized reward, holding the state terminal and setting rewards to zero after termination. For any control law, condition on its history before each action and apply the inequality to that chosen action. Induction, multiplying the step- t inequality by γ^t and cancelling successive continuation terms, gives

$$\mathbb{E} \left[\sum_{t=0}^{n-1} \gamma^t r_t + \gamma^n v_\pi(s_n) \mid s_0 = s \right] \leq v_\pi(s),$$

where the value at a terminal state is zero. The final continuation term has absolute expectation at most $\|v_\pi\|_\infty \gamma^n$ in the discounted case, or $\|v_\pi\|_\infty \mathbb{P}(T > n)$ in the undiscounted case. The remainder tends to zero, using Lemma 1 in the latter case. Finite expected absolute rewards and the same tail bounds justify passing to the full return. Policy π attains v_π , so it is optimal.

Finally choose π maximizing $\sum_s v_\pi(s)$ over the finite set $\mathcal{P}$. If it were not greedy for q_π , a greedy policy π' would make $h \geq 0$ with a positive coordinate. The $j = 0$ term in (4) would strictly increase the sum, a contradiction. All optimal policies share the values denoted by v_{π_*} and q_{π_*} . ■

1.2 Initial-visit updates and information

Let $\mathcal{Q} = \mathbb{R}^{\mathcal{I}}$. For $q \in \mathcal{Q}$, retain the manuscript's notation

$$P(q) = \{\pi \in \mathcal{P} : q(s, \pi(s)) = \max_a q(s, a) \text{ for every } s\}, \quad Q(\pi) = \{q \in \mathcal{Q} : \pi \in P(q)\}.$$

MC-O-PI starts from a finite deterministic table q_0 . At iteration k it selects $\pi_k \in P(q_k)$, draws the initial pair I_{k+1} according to σ_k , obtains a return G_{k+1} by following π_k after that pair, and updates only this pair:

$$q_{k+1}(i) = q_k(i) + \alpha_k u_k(i) (q_{\pi_k}(i) + v_k(i) - q_k(i)), \quad i \in \mathcal{I}, \quad (5)$$

where $u_k(i) = \mathbf{1}_{\{I_{k+1}=i\}}$. On the sampled pair, $v_k(i) = G_{k+1} - q_{\pi_k}(i)$; set $v_k(i) = 0$ otherwise. For ordinary episodic returns the policy remains fixed until termination.

The history $\mathcal{F}_k$ contains the information available after q_k is computed but before selecting π_k and σ_k . Write $\mathcal{F}'_k = \sigma(\mathcal{F}_k, \pi_k, \sigma_k)$ for the post-selection history. Thus

$$\mathcal{F}_k \subseteq \mathcal{F}'_k \subseteq \sigma(\mathcal{F}'_k, I_{k+1}) \subseteq \mathcal{F}_{k+1}.$$

A history is represented mathematically by a sigma-algebra, and conditional expectation averages over the remaining randomness given that information. An increasing sequence of histories is a filtration. All stopping times below refer to $(\mathcal{F}'_k)$: whether such a time has occurred by k can be decided from $\mathcal{F}'_k$. For example, the first index at which $\|q_k\|_\infty > K$ is a stopping time, since the tables already observed determine whether a crossing has occurred. We use the tower property: averaging a conditional expectation again over a smaller history gives the conditional expectation for that smaller history.

Assumption 3 (Standing assumptions) *The deterministic steps satisfy $0 < \alpha_k \leq 1$ and $\sum_k \alpha_k^2 < \infty$. The sampling law may depend on the entire available history and the selected policy, but every pair receives infinite conditional learning weight:*

$$\sum_{k=0}^{\infty} \alpha_k \sigma_k(i) = \infty \quad \text{almost surely for every } i \in \mathcal{I}. \quad (6)$$

For a deterministic constant $c_v < \infty$, almost surely on $\{\sigma_k(i) > 0\}$,

$$\mathbb{E}[G_{k+1} \mid \mathcal{F}'_k, I_{k+1} = i] = q_{\pi_k}(i), \quad \mathbb{E}[(G_{k+1} - q_{\pi_k}(i))^2 \mid \mathcal{F}'_k, I_{k+1} = i] \leq c_v. \quad (7)$$

No restriction is imposed on a conditional version where $\sigma_k(i) = 0$.

Condition (6) implies $\sum_k \alpha_k = \infty$, giving the usual Robbins–Monro conditions. Infinite visitation alone would not suffice: infinitely many updates can still have finite total learning weight. Bounded one-step rewards ensure (7) for ordinary episode returns, by discounting or Lemma 1; otherwise we explicitly assume these return moments.

For initial visits, the update distribution is $\mu_k(i) = \mathbb{E}[u_k(i) \mid \mathcal{F}'_k] = \sigma_k(i)$. The combined noise and mean-field representation are

$$w_k(i) = u_k(i)v_k(i) + (u_k(i) - \mu_k(i))(q_{\pi_k}(i) - q_k(i)), \quad (8)$$

$$q_{k+1} = q_k + \alpha_k (\mu_k(q_{\pi_k} - q_k) + w_k), \quad (9)$$

with coordinatewise products. The moments in (7) imply

$$\mathbb{E}[w_k(i) \mid \mathcal{F}'_k] = 0, \quad \mathbb{E}[w_k(i)^2 \mid \mathcal{F}'_k] \leq \sigma_k(i) \{c_v + (B + \|q_k\|_\infty)^2\}. \quad (10)$$

Indeed, $w_k(i)$ is the centered version of $u_k(i)(G_{k+1} - q_k(i))$, and variance is bounded by its second moment. Fixed-time tables have finite second moments, by induction in (5).

1.3 Accumulated noise and persistent targets

A martingale difference is a random increment whose conditional mean given the previous history is zero. We use the following elementary consequence of square summability: if such increments e_{k+1} satisfy $\sum_k \mathbb{E}[e_{k+1}^2] < \infty$, their partial sums converge almost surely. To see the tail control behind this fact, orthogonality of distinct increments and the martingale maximal inequality give

$$\mathbb{P} \left(\sup_{n \geq r} \left| \sum_{k=r}^{n-1} e_{k+1} \right| > \varepsilon \right) \leq \varepsilon^{-2} \sum_{k=r}^{\infty} \mathbb{E}[e_{k+1}^2].$$

Here is a direct finite-horizon justification. Put $S_n = \sum_{k=r}^{n-1} e_{k+1}$ and fix a terminal index N . Distinct increments have zero expected product: for $j < k$, conditioning $e_{j+1}e_{k+1}$ on the history at k gives zero. Consequently $\mathbb{E}[S_N^2] = \sum_{k=r}^{N-1} \mathbb{E}[e_{k+1}^2]$. Let θ be the first index with $|S_\theta| > \varepsilon$. On each event $\{\theta = j\}$, which is known at j , the later sum $S_N - S_j$ has conditional mean zero. Expanding the square therefore gives

$$\mathbb{E}[\mathbf{1}_{\{\theta=j\}} S_N^2] \geq \mathbb{E}[\mathbf{1}_{\{\theta=j\}} S_j^2] \geq \varepsilon^2 \mathbb{P}(\theta = j).$$

Sum over $j \leq N$ and then let $N \rightarrow \infty$ to obtain the displayed maximal inequality. Choose r_j making the variance tail at most 2^{-3j} and take $\varepsilon = 2^{-j}$; the probabilities then sum to a finite number. The elementary Borel–Cantelli implication (events whose probabilities sum to a finite number occur only finitely often) shows that these tails are uniformly small eventually. Hence the partial sums are Cauchy almost surely.

Lemma 4 (Actual learning weight) *Under the standing assumptions, almost surely*

$$\sum_k \alpha_k u_k(i) = \infty \quad (i \in \mathcal{I}).$$

Proof. The increments $\alpha_k(u_k(i) - \sigma_k(i))$ have conditional mean zero given $\mathcal{F}'_k$ and second moment at most α_k^2 . Their series converges by the preceding fact. Thus the partial sums of actual and conditional learning weights differ by a convergent sequence. Equation (6) proves the claim. Intersect the probability-one events over the finitely many pairs. $\blacksquare$

Lemma 5 (Filtering accumulated errors) *Consider a scalar recursion $x_{k+1} = (1 - b_k)x_k + b_k y_k + e_{k+1}$ with $0 \leq b_k \leq 1$. For $n > r$,*

$$\left| \sum_{k=r}^{n-1} e_{k+1} \prod_{j=k+1}^{n-1} (1 - b_j) \right| \leq 2 \max_{r \leq m \leq n} \left| \sum_{k=r}^{m-1} e_{k+1} \right|. \quad (11)$$

If $\sum_k b_k = \infty$ and $\sum_k e_{k+1}$ converges, eventual bounds $\ell \leq y_k \leq h$ imply $\ell \leq \liminf_k x_k \leq \limsup_k x_k \leq h$. In particular, an eventually constant target y gives $x_k \rightarrow y$. These are pathwise statements, with no independence assumption.

Proof. Unrolling the recursion assigns the initial value weight $\prod_{k=r}^{n-1} (1 - b_k)$, the targets nonnegative weights summing to one minus this product, and the errors the weights in (11). Put $E_m = \sum_{k=r}^{m-1} e_{k+1}$ and $a_k = \prod_{j=k+1}^{n-1} (1 - b_j)$. Summation by parts gives

$$\sum_{k=r}^{n-1} (E_{k+1} - E_k) a_k = E_n a_{n-1} - \sum_{k=r+1}^{n-1} E_k (a_k - a_{k-1}).$$

Here $0 \leq a_r \leq \dots \leq a_{n-1} = 1$, proving (11). Since $\sum_k b_k = \infty$, the initial weight tends to zero for fixed r . Convergence of the error series makes its tail partial sums uniformly small. Take r after the target bounds begin, let $n \rightarrow \infty$, and then let $r \rightarrow \infty$. ■

Lemma 6 (Stability and persistent targets) *Under the standing assumptions, on one probability-one event,*

$$\limsup_k \|q_k\|_\infty \leq B.$$

On the same event, if $q_{\pi_k} = \bar{q}$ for all sufficiently large k , then $q_k \rightarrow \bar{q} = q_{\pi_}$.*

Proof. For a pair i , use $b_k = \alpha_k u_k(i)$, $y_k = q_{\pi_k}(i)$ and $e_{k+1} = \alpha_k u_k(i) v_k(i)$ in Lemma 5. The weights sum to infinity by Lemma 4. The error increments have conditional mean zero and second moment at most $c_v \alpha_k^2$, so their series converges almost surely. Intersect these events over all pairs. The targets lie in $[-B, B]$, giving the asserted bound, and an eventual target $\bar{q}$ gives $q_k \rightarrow \bar{q}$.

Choose a policy $\hat{\pi}$ occurring infinitely often after the target has become constant. It satisfies $q_{\hat{\pi}} = \bar{q}$. Along that subsequence its greedy inequalities pass to the limit, so it is greedy for its own action values. Policy improvement and verification therefore give $\bar{q} = q_{\pi_*}$. This argument is pathwise; it does not condition on the event that the target eventually becomes constant. ■

A transition time records a change of the target, not merely a change of policy label. Starting at $t_0 = 0$, define

$$t_{n+1} = \inf\{k > t_n : q_{\pi_k} \neq q_{\pi_{t_n}}\} \quad (t_n < \infty), \quad (12)$$

with $\inf \emptyset = \infty$ and $t_{n+1} = \infty$ when $t_n = \infty$. These are stopping times for $(\mathcal{F}'_k)$. On $[t_n, t_{n+1})$ the target is constant even if the selected policy changes. To prove convergence it therefore suffices to prove that only finitely many transitions occur. A nonoptimal policy can share the optimal action-value table, so its selection alone does not imply a future target change.

2 Mean-field policy improvement

Averaging the sampled update over its pair and return gives

$$q_{k+1}(s, a) = q_k(s, a) + \alpha_k \mu_k(s, a) (q_{\pi_k}(s, a) - q_k(s, a)), \quad \pi_k \in P(q_k). \quad (13)$$

For this deterministic recursion, impose $0 < \alpha_k \leq 1$, $0 \leq \mu_k(s, a) \leq 1$ and uniform updates within each state:

$$\mu_k(s, a) = \mu_k(s) \quad (a \in \mathcal{A}(s)), \quad \sum_k \alpha_k \mu_k(s) = \infty \quad (s \in \mathcal{S}). \quad (14)$$

The scalar $\mu_k(s)$ is the common coefficient for actions at s . When derived from a sampling law, it is a per-pair probability, not the state marginal. Use full inertia: choose π_0 greedily, and thereafter

$$\pi_{k+1}(s) = \pi_k(s) \quad \text{whenever } \pi_k(s) \in \arg \max_a q_{k+1}(s, a). \quad (15)$$

If the preceding action is no longer greedy, any maximizing action may be selected.

Proposition 7 *Under these conditions $v_{\pi_{k+1}} \geq v_{\pi_k}$ and $q_{\pi_{k+1}} \geq q_{\pi_k}$ for every k . At every finite target transition the state values improve strictly in at least one state.*

Proof. Fix s and put $c = \alpha_k \mu_k(s)$. If $c > 0$, greediness before and after the update gives

$$q_{\pi_k}(s, \pi_{k+1}(s)) - q_{\pi_k}(s, \pi_k(s)) \geq \frac{1-c}{c} (q_k(s, \pi_k(s)) - q_k(s, \pi_{k+1}(s))) \geq 0.$$

If $c = 0$, the table at s is unchanged and full inertia retains the action, giving equality. Lemma 2 proves both comparisons. Across a target transition they remain valid by transitivity, and distinct action-value targets force a strict state-value comparison. ■

Proposition 8 *Under the same conditions, $q_k \rightarrow q_{\pi^*}$. This conclusion needs neither square summability nor step-size regularity.*

Proof. The target sequence (q_{π_k}) is nondecreasing and takes values in the finite set $\{q_\pi : \pi \in \mathcal{P}\}$, so it is eventually a single target $\bar{q}$. Apply Lemma 5 to each coordinate, with $b_k = \alpha_k \mu_k(s)$ and zero errors, to obtain $q_k \rightarrow \bar{q}$. A policy occurring infinitely often is greedy for $\bar{q} = q_\pi$, and hence optimal by Lemma 2. ■

The common coefficient c is the reason for requiring equal action rates within each state. Different states may have different rates and may pause arbitrarily long, provided their cumulative learning diverges. Fixed-priority ties give the same deterministic result: an unchanged table retains the same priority maximizer, which is the only tie property used above.

3 Stochastic convergence under cumulative learning

In the sampled algorithm, a single update can worsen the policy even though the mean field improves it. We show instead that each sufficiently late interval with a fixed target has a uniformly positive chance of ending in an improvement or of continuing forever. Boundedness lets us prove this statement before leaving a fixed ball. Finitely many target values then rule out infinitely many unsuccessful transitions.

In addition to the standing assumptions, impose full inertia (15), and sample the initial action uniformly within its state:

$$\sigma_k(s, a) = \sigma_k(s) \geq 0, \quad \sum_s |\mathcal{A}(s)| \sigma_k(s) = 1. \quad (16)$$

Here $\sigma_k(s)$ is the probability of each pair at s ; the state marginal is $|\mathcal{A}(s)| \sigma_k(s)$. Thus (6) reduces to $\sum_k \alpha_k \sigma_k(s) = \infty$ at every state. We also require the following weaker form of eventual nonincrease.

Assumption 9 (Bounded future increases) *There are deterministic $C_\alpha \geq 1$ and $k_\alpha < \infty$ such that*

$$\alpha_j \leq C_\alpha \alpha_k \quad (j \geq k \geq k_\alpha). \quad (17)$$

Eventual nonincrease is the case $C_\alpha = 1$. The condition also permits infinitely many increases, for example $\alpha_k = (2 + (-1)^k)/(k + 3)$ with $C_\alpha = 3$. It imposes no lower bound on future steps relative to the current one.

Theorem 10 *Under the standing assumptions, initial-visit updates, uniform initial actions (16), full inertia (15) and bounded future step increases (17), almost surely (q_{π_k}) changes only finitely many times and*

$$q_k \longrightarrow q_{\pi_*}.$$

Every sufficiently late selected policy is optimal. If every state has a unique optimal action, π_k is eventually constant.

The proof occupies the remainder of this section. The following probability estimate controls an estimated gap between two actions. Its dependence on a state's own accumulated learning, rather than elapsed iteration count, allows arbitrarily long waits between updates of that state.

4 A barrier on a state's learning clock

The same scalar weight appears in a gap's drift and variance. Keeping this weight allows arbitrarily long pauses. When it is zero, the noise vanishes and the drift cannot decrease the gap. No divergence of the accumulated weight is needed for the following local estimate.

Lemma 11 (State-clock barrier) *Let D_k be real and $\mathcal{F}'_k$ -measurable, with integrable increments. Let $r \leq \tau$ be stopping times, and let $\sigma_k(s) \in [0, 1]$ be an $\mathcal{F}'_k$ -measurable scalar weight for a fixed state s . In the application it will be the probability of each initial pair at s . The deterministic steps are in $(0, 1]$, square summable, and satisfy $\alpha_j \leq C_\alpha \alpha_k$ for all $j \geq k \geq k_\alpha$, with deterministic $C_\alpha \geq 1$ and k_α . Assume $r \geq k_\alpha$. Write*

$$m_k = \mathbb{E}[D_{k+1} - D_k \mid \mathcal{F}'_k], \quad Z_{k+1} = D_{k+1} - D_k - m_k.$$

Suppose deterministic $\delta, V > 0$ satisfy, on $\{r \leq k < \tau, D_k \geq 0\}$,

$$m_k \geq \alpha_k \sigma_k(s) (4\delta - D_k), \quad \mathbb{E}[Z_{k+1}^2 \mid \mathcal{F}'_k] \leq V \alpha_k^2 \sigma_k(s). \quad (18)$$

Then, on $\{r < \infty, D_r > 0\}$,

$$\mathbb{P}(\exists \text{ finite } n \in [r, \tau] : D_n \leq 0 \mid \mathcal{F}'_r) \leq C_0 \frac{C_\alpha \alpha_r}{\min\{D_r, \delta\}} + \frac{C_1}{\delta^2} \sum_{k=r}^{\infty} \alpha_k^2, \quad (19)$$

where $C_0 = \max\{1, 20V(1 + (3\delta)^{-1})\}$ and $C_1 = 16V$. The crossing event includes the update ending at τ .

Proof. Before reaching δ . Put $a_r = C_\alpha \alpha_r$, so every future step is at most a_r . Suppose $0 < D_r < \delta$. If $D_r < a_r$, the first term in (19) is already at least one. Otherwise define

$$\nu = \tau \wedge \inf\{n \geq r : D_n \leq 0 \text{ or } D_n \geq \delta\}, \quad M_n = \sum_{k=r}^{n-1} \mathbf{1}_{\{k < \nu\}} Z_{k+1},$$

and the state clock

$$A_s(r, n) = \sum_{k=r}^{n-1} \alpha_k \sigma_k(s). \quad (20)$$

A first crossing at $n \leq \tau$ before reaching δ forces

$$M_n \leq -D_r - 3\delta A_s(r, n). \quad (21)$$

For $b > 0$ known at r , set $T(b) = \inf\{n \geq r : A_s(r, n) \geq b\}$, possibly infinite. The bound $\alpha_k \leq a_r$ controls the clock overshoot and future variance, at every finite horizon $n \geq r$, by

$$A_s(r, T(b) \wedge n) \leq b + a_r, \quad \sum_{k=r}^{T(b) \wedge n-1} \alpha_k^2 \sigma_k(s) \leq a_r(b + a_r).$$

To justify the conditional estimate, first fix a deterministic restart r and stop at $T(b) \wedge N$ for a finite horizon N . Stopping preserves zero conditional means because whether an increment is retained is known before that increment is drawn. Orthogonality and the variance bound give

$$\mathbb{E}[M_{T(b) \wedge N}^2 \mid \mathcal{F}'_r] \leq V a_r(b + a_r).$$

The threshold-crossing proof given in Section 1 works conditionally too: multiply its expectations by the indicator of any event known at r . The tower property then gives the same inequality conditional on $\mathcal{F}'_r$. Quantities b, z known at r remain fixed in this calculation. In particular, the following bound holds (even the constant 1 in place of 4 suffices):

$$\mathbb{P}\left(\sup_{r \leq n \leq T(b)} |M_n| \geq z \mid \mathcal{F}'_r\right) \leq \frac{4V a_r(b + a_r)}{z^2}, \quad z > 0. \quad (22)$$

The supremum is over finite indices. First stop at a finite horizon, then increase the horizon; an unattained supremum is handled by increasing thresholds to z from below. For possibly unbounded $\mathcal{F}'_r$ -measurable b, z , perform the calculation on $\{b \leq m, 1/m \leq z \leq m\}$ and increase m ; these events are known at r and exhaust $\{b, z > 0\}$. Random r is treated on

$\{r = j\}$ and then combined over integers j . Thus no finite bound on $T(b)$ or the waiting time has been assumed.

Partition possible crossing times into $A_s(r, n) < D_r/(3\delta)$ and, for $j \geq 1$,

$$2^{j-1}D_r/(3\delta) \leq A_s(r, n) < 2^j D_r/(3\delta).$$

Apply (22) with $(b, z) = (D_r/(3\delta), D_r)$ for the first range and $(b, z) = (2^j D_r/(3\delta), 2^{j-1} D_r)$ for the others. Because $D_r \geq a_r$, the respective bounds are at most

$$4V(1 + (3\delta)^{-1}) \frac{a_r}{D_r}, \quad 16V(1 + (3\delta)^{-1}) \frac{a_r}{D_r} 2^{-j}.$$

Summing and using $a_r = C_\alpha \alpha_r$ gives the first term in (19).

After reaching δ . Let $\rho = \inf\{n \geq r : D_n \geq \delta\}$, using $\rho = r$ if $D_r \geq \delta$. On $\{\rho < \infty, \rho \leq \tau\}$, restart at ρ and stop at $\eta = \tau \wedge \inf\{n \geq \rho : D_n \leq 0\}$:

$$\widetilde{M}_n = \sum_{k=\rho}^{n-1} \mathbf{1}_{\{k < \eta\}} Z_{k+1}.$$

For $D_k \geq \delta$, (18) and $0 \leq \alpha_k \sigma_k(s) \leq 1$ imply $D_k + m_k \geq \delta$. If a first subsequent crossing is at n , let $\ell < n$ be the last index with $D_\ell \geq \delta$. All subsequent gaps before n are in $(0, \delta)$ and have nonnegative mean increments, so

$$D_n \geq \delta + \widetilde{M}_n - \widetilde{M}_\ell.$$

A crossing forces $\sup_{n \geq \rho} |\widetilde{M}_n| \geq \delta/2$. Doob's inequality at ρ bounds this probability by $16V\delta^{-2} \sum_{k \geq \rho} \alpha_k^2$. Multiply by the $\mathcal{F}'_\rho$ -measurable event $\{\rho < \infty, \rho \leq \tau\}$, average back to $\mathcal{F}'_r$, and use $\rho \geq r$. This proves the second term. The last-visit index ℓ is used only in a pathwise inequality; we never condition at it. Both parts retain the increment $k = \tau - 1$ and therefore protect the gap through the endpoint itself. $\blacksquare$

4.1 Reference gaps and separately completed seeds

For a reference policy π , define

$$\begin{aligned} d_\pi(s, a) &= q_\pi(s, \pi(s)) - q_\pi(s, a), & \mathcal{B}_\pi &= \{(s, a) : d_\pi(s, a) > 0\}, \\ d_k(s, a) &= q_k(s, \pi(s)) - q_k(s, a). \end{aligned} \tag{23}$$

An action in $\mathcal{B}_\pi$ would worsen the reference action according to q_π . If these estimated gaps are all positive, no greedy policy can select a bad action, so policy improvement applies. If every $\mathcal{B}_\pi$ is empty, improvement applies between every pair of policies in both directions. Their state values are equal; choosing a policy taking any prescribed action shows that every action attains that same value. All policies are optimal with the same target, and Lemma 6 completes the theorem. Otherwise set

$$d_* = \min_{\pi, (s, a) \in \mathcal{B}_\pi} d_\pi(s, a) > 0, \quad \delta = d_*/4, \quad h = g = \delta/2. \tag{24}$$

Fix deterministic $K > B$ and restart r , and let $\zeta_r = \inf\{k \geq r : \|q_k\|_\infty > K\}$. For a stopping time $t \geq r$, work on $\{t < \infty, t < \zeta_r\}$, fix $\pi = \pi_t$, and define

$$\tau = \inf\{k > t : q_{\pi_k} \neq q_\pi\}, \quad \chi = \tau \wedge \zeta_r. \quad (25)$$

These are stopping times. Return moments give finite second moments for every fixed-time table by induction in (5). For $t \leq k < \chi$, the action-value target is exactly q_π , even if the selected policy label is different. Equal initial action probabilities give

$$\mathbb{E}[d_{k+1}(s, a) - d_k(s, a) \mid \mathcal{F}'_k] = \alpha_k \sigma_k(s)(d_\pi(s, a) - d_k(s, a)), \quad (26)$$

$$\text{Var}(d_{k+1}(s, a) - d_k(s, a) \mid \mathcal{F}'_k) \leq V \alpha_k^2 \sigma_k(s), \quad V = 2\{c_v + (B + K)^2\}. \quad (27)$$

Only the two coordinates forming the gap can change, their selection events are disjoint, and each contributes at most $\alpha_k^2 \sigma_k(s)\{c_v + (B + K)^2\}$ to the raw second moment. For each bad pair, $d_\pi(s, a) \geq 4\delta$, so the barrier's drift and variance hypotheses hold through χ . Its probability bound will be used once a positive starting gap has been obtained. All such moment estimates are used before exit, not conditioned on future containment in the ball.

A favorable local update. Suppose $\pi_k(s) = \pi(s)$ and a current bad gap is below δ . Request one of the following action/return events:

1. If $q_\pi(s, \pi(s)) - q_k(s, \pi(s)) \geq d_*/2$, select $\pi(s)$ and require $G_{k+1} \geq q_\pi(s, \pi(s)) - h$. The reference coordinate increases by at least $3\delta\alpha_k/2$.
2. Otherwise select a and require $G_{k+1} \leq q_\pi(s, a) + h$. The identity

$$q_k(s, a) - q_\pi(s, a) = q_k(s, \pi(s)) - q_\pi(s, \pi(s)) + d_\pi(s, a) - d_k(s, a) > \delta$$

implies that the bad coordinate decreases by more than $g\alpha_k = \delta\alpha_k/2$.

In either case no other gap against the reference action decreases, and the reference action remains greedy. For centered X with variance at most c_v , the one-sided Chebyshev inequality gives

$$\mathbb{P}(X \geq -h) \geq \frac{h^2}{c_v + h^2}, \quad \mathbb{P}(X \leq h) \geq \frac{h^2}{c_v + h^2}.$$

For example, apply Markov's inequality to $(X + c_v/h)^2$ to bound $\mathbb{P}(X \geq h)$ and repeat for $-X$; when $c_v = 0$, $X = 0$ almost surely. Conditional on selecting s , its initial action is uniform, so every requested local update has conditional probability at least

$$p_0 = \frac{h^2}{\max_s |\mathcal{A}(s)| (c_v + h^2)} > 0. \quad (28)$$

No probability of selecting the state enters this bound.

Lemma 12 (Separately completed statewise seeds) *For full-inertia ties and any positive integer N , there is an adapted construction asking for at most JN favorable local updates, starting at $t \geq k_\alpha$. Along successful requests, every unfinished state retains its reference action, including at an endpoint reached by a favorable update. If state s completes at time $T_s < \infty$, then*

$$d_{T_s}(s, a) \geq \min\{\delta, gN\alpha_{T_s}/C_\alpha\} \quad ((s, a) \in \mathcal{B}_\pi). \quad (29)$$

Each request, conditional on selecting its unfinished state, has probability at least p_0 .

Proof. Order the bad actions at each state. Skip an action immediately if its gap is already at least δ ; otherwise process it with at most N favorable updates at that state, stopping earlier if it reaches δ . Process skips at t and after every favorable update, before the next sample. A state completes when its list is exhausted. Write T_s for this time, using $T_s = t$ for a state completed initially and $T_s = \infty$ if the construction stops before completion. These are stopping times. Every selection of an unfinished state requests a local update; completed states receive ordinary updates. Stop the construction at the first failed requirement or at χ .

Under full inertia, raising the reference estimate or lowering a competitor retains the reference action, since it remains greedy. At an unupdated state the table is unchanged, so the action is also retained. Induction proves the assertion through every unfinished state's wait.

The same retention conclusion also holds for fixed priorities: at time t the reference action has highest priority among that state's maximizers. Leaving the table unchanged, raising its estimate, or lowering a competitor cannot create a new maximizer ahead of it in the priority order. Therefore that same action continues to be selected at every unfinished state. Induct over all iterations, including those updating other states and the final favorable update at completion. This proves the required retention property directly; a fixed-priority rule is not being identified with full inertia on arbitrary updates.

Every processed gap either reaches δ or receives N increases of at least $g\alpha_k$. Later seed updates at its state cannot lower it. Bounded future increases give $\alpha_k \geq \alpha_{T_s}/C_\alpha$ for every such update and proves (29). At most J bad actions are processed, so there are at most JN requests, irrespective of waiting times. Their conditional probability bound is (28). ■

Completed states can be updated while other states wait. We therefore cannot condition indiscriminately on all future successful behavior: the completed-state return laws must stay unchanged. The following bounded change of measure conditions only the seed updates.

An auxiliary probability law is a way of reweighting possible runs for the proof. If a nonnegative random weight L_∞ has expectation one, define

$$\hat{\mathbb{P}}(E) = \mathbb{E}[L_\infty \mathbf{1}_E]$$

for every event E . Runs with larger weight become more likely under this law, and an event of original probability zero still has probability zero. This last property is denoted by $\hat{\mathbb{P}} \ll \mathbb{P}$. We choose the weights so that requested updates always succeed, while the laws at completed states remain unchanged.

Lemma 13 (Conditioning only the seed updates) *There is an auxiliary law $\hat{\mathbb{P}} \ll \mathbb{P}$ under which all active requirements of Lemma 12 succeed. Conditional state selection is unchanged. At completed states, and after the construction stops, the conditional action and return law given the current history and selected state is unchanged. For every future event E , on $\{t < \infty, t < \zeta_r\}$,*

$$\mathbb{P}(E \mid \mathcal{F}'_t) \geq p_0^{JN} \hat{\mathbb{P}}(E \mid \mathcal{F}'_t). \quad (30)$$

Proof. Let S_{k+1} be the state component of I_{k+1} . At an active unfinished state s , let $p_k(s)$ be the conditional probability of its requirement given $\mathcal{F}'_k, S_{k+1} = s$. Then $p_k(s) \geq p_0$. If the state has zero sampling probability, set $p_k(s) = p_0$; this value is never used on a positive-probability draw.

Start $L_t = 1$. At a requested update of s , multiply L by the indicator of success divided by $p_k(s)$; at other iterations multiply it by one. Every factor has conditional mean one given $\mathcal{F}'_k, S_{k+1}$. Hence (L_k) is a martingale. At most JN factors differ from one, even though their times need not be bounded, so $0 \leq L_k \leq p_0^{-JN}$. Since only finitely many factors differ from one, L_k is eventually constant on each path. Bounded convergence gives $\mathbb{E}[L_\infty | \mathcal{F}'_t] = 1$. Define $\widehat{\mathbb{P}}(E) = \mathbb{E}[L_\infty \mathbf{1}_E]$ as above. For random t , use likelihood one before t , on $\{t = \infty\}$, and off $\{t < \zeta_r\}$, and combine the constructions on $\{t = j\}$ over integers j . Each such event is known at j , so the fixed-start conditional calculation applies there. The law agrees with $\mathbb{P}$ on $\mathcal{F}'_t$ where the construction is active.

A failure makes $L_\infty = 0$, so active requirements succeed almost surely under the auxiliary law. Here is why later requests do not bias a completed-state update. For any bounded random variable Y determined by the next history, the definition of weighted probabilities and the tower property give, on $\{L_k > 0\}$,

$$\widehat{\mathbb{E}}[Y | \mathcal{F}'_k] = \frac{\mathbb{E}[L_\infty Y | \mathcal{F}'_k]}{L_k} = \frac{\mathbb{E}[L_{k+1} Y | \mathcal{F}'_k]}{L_k}.$$

The second equality uses $\mathbb{E}[L_\infty | \mathcal{F}'_{k+1}] = L_{k+1}$, so future factors have averaged out. Write $R_{k+1} = L_{k+1}/L_k$ for the current factor. Since $\mathbb{E}[R_{k+1} | \mathcal{F}'_k, S_{k+1}] = 1$, applying the formula to state indicators preserves the state-selection probabilities. Conditioning further on $S_{k+1} = s$ gives

$$\widehat{\mathbb{E}}[Y | \mathcal{F}'_k, S_{k+1} = s] = \mathbb{E}[R_{k+1} Y | \mathcal{F}'_k, S_{k+1} = s].$$

At a completed state this factor is one, preserving its conditional action and return law. The same holds after the construction stops. Histories where $L_k = 0$ have auxiliary probability zero, so no division by zero is needed. Absolute continuity preserves the almost-sure algorithmic identities, including the full-inertia tie rule. Finally,

$$\widehat{\mathbb{P}}(E | \mathcal{F}'_t) = \mathbb{E}[L_\infty \mathbf{1}_E | \mathcal{F}'_t] \leq p_0^{-JN} \mathbb{P}(E | \mathcal{F}'_t),$$

which proves the bound. This auxiliary law is used only in the proof; the algorithm still samples its original returns. $\blacksquare$

4.2 A stopped positive probability of progress

Let $W(\pi) = \sum_{i \in \mathcal{I}} q_\pi(i)$. Its possible values form a finite ordered set. A distinct componentwise larger target has strictly larger W , although different targets may have equal W .

Lemma 14 (Stopped improvement bound) *Under the assumptions of Theorem 10, for every fixed $K > B$ there are deterministic k_0 and $p > 0$ such that for every deterministic restart r and stopping time $t \geq \max\{r, k_0\}$, on $\{t < \infty, t < \zeta_r\}$,*

$$\mathbb{P}\left(\left\{\zeta_r \leq \tau, \zeta_r < \infty\right\} \text{ or } \left\{\tau = \zeta_r = \infty\right\} \text{ or } \left\{\tau < \zeta_r, W(\pi_\tau) > W(\pi_t)\right\} \middle| \mathcal{F}'_t\right) \geq p. \quad (31)$$

Here τ is the next target change in (25).

Proof. Fix V from (27) and C_0, C_1 from the state-clock barrier. Choose a positive integer N with $2JC_0C_\alpha^2/(\delta N) \leq 1/4$. Then choose deterministic $k_0 \geq k_\alpha$ so that, for every $t \geq k_0$,

$$N\alpha_t \leq 2C_\alpha, \quad \frac{JC_1}{\delta^2} \sum_{k=t}^{\infty} \alpha_k^2 \leq \frac{1}{4}. \quad (32)$$

The radius and variance constants are fixed before choosing N .

Use the auxiliary law of Lemma 13. For each bad action whose state completes at $T_s < \chi$, apply the barrier at T_s , stopped at χ . At that completed state the original sampling and return laws are preserved, so (26)–(27) hold under $\mathbb{P}$. No such assertion is made for a conditioned return at an unfinished state. Bounded future increases and (29)–(32) imply that its conditional crossing probability is at most

$$\frac{2C_0C_\alpha^2}{\delta N} + \frac{C_1}{\delta^2} \sum_{k=t}^{\infty} \alpha_k^2. \quad (33)$$

Indeed the choice of k_0 applies to every $T_s \geq t$, so $gN\alpha_{T_s}/C_\alpha \leq \delta$ and

$$\frac{C_\alpha \alpha_{T_s}}{\min\{d_{T_s}(s, a), \delta\}} \leq \frac{C_\alpha^2}{gN}.$$

Multiply the conditional estimate by the $\mathcal{F}'_{T_s}$ -measurable event $\{T_s < \chi\}$ and average back to $\mathcal{F}'_t$. A union bound over at most J pairs gives probability at most $1/2$ of any completed gap failing through χ . This needs neither independence nor simultaneous completion. If $T_s = \chi < \infty$, its final favorable update already retains the reference action, so no subsequent barrier is needed.

On the complementary event, either exit occurs first, the target persists without exit, or $\tau < \zeta_r$ is finite. In the third case, states unfinished just before the endpoint retain the reference action, including those completed at the endpoint. Previously completed states cannot choose a bad action. Thus

$$q_\pi(s, \pi_\tau(s)) \geq q_\pi(s, \pi(s)) \quad \text{for every } s.$$

Policy improvement gives $q_{\pi_\tau} \geq q_\pi$ with distinct targets, and hence a strict increase of W . There is no need to prove that every seed completes: at a finite endpoint unfinished states are safe, and an infinite endpoint without exit is already a successful outcome. Transfer back by (30), obtaining (31) with $p = \frac{1}{2}p_0^{JN}$.

If $\mathcal{B}_\pi$ is empty, the gap conditions and barrier are vacuous and policy improvement holds for every next selected policy. The same lower bound remains valid. An exit *after* an unsuccessful finite target change is not counted as success for that earlier interval. ■

4.3 Finitely many target changes and convergence

Proof of Theorem 10. Fix $K > B$ and constants k_0, p from Lemma 14. For a deterministic $r \geq k_0$, start at $t_0 = r$ and absorb immediately if $\|q_r\|_\infty > K$. Enumerate successive target changes

$$t_{n+1} = \inf\{k > t_n : q_{\pi_k} \neq q_{\pi_{t_n}}\}$$

only while $t_n < \zeta_r$. All finite continuing times are stopping times. Stop counting if exit occurs before or at the next target change, or if the target persists forever without exit.

At every alive interval, absorption or a strict increase in W has conditional probability at least p . By induction and the tower property, absorption within the next m intervals or m consecutive strict increases has conditional probability at least p^m . On an absorbed branch the continuation probability is one; on a finite successful-transition branch condition at its stopping time and apply the remaining bound. This uses no independence and never conditions at an infinite time on a continuing branch.

Let L be the number of distinct values of W . There cannot be L consecutive strict increases. Absorption within the next L intervals therefore has conditional probability at least p^L . Applying the bound at successive surviving blocks gives probability at most $(1 - p^L)^j$ of surviving j such blocks. In particular the probability of infinitely many target changes before exit is zero.

Intersect these probability-one conclusions over all deterministic integers $r \geq k_0$. Lemma 6 says that almost every path remains inside the ball after some such integer. On that path, choose r with $\zeta_r = \infty$; the original sequence consequently has only finitely many target changes. The integer is selected pathwise only after the countable intersection. We do not condition on an eventual containment time or claim that it is a stopping time. Nor is an unconditional geometric tail for the original process's total number of changes asserted.

Lemma 6 now gives $q_k \rightarrow q_{\pi_*}$. Every nonoptimal action has a strictly positive optimality gap. Finiteness of the action sets and convergence exclude all such actions from sufficiently late greedy policies. Those policies are greedy for q_{π_*} and hence optimal by Bellman verification. If every state has a unique optimal action, their labels are eventually constant. ■